METRIC
FF-B-171/21
30 NOVEMBER 1993

FEDERAL SPECIFICATION SHEET

BEARING, BALL, ANNULAR, SINGLE ROW, ANGULAR CONTACT, COUNTERBORED OUTER RING, CONTACT ANGLE ABOVE 10° UP TO AND INCLUDING 22°, NON-SEPARABLE, ANSI/AFBMA DIMENSION SERIES 02.

The complete requirements for acquiring the bearings described herein shall consist of this document and the latest issue of FF-B-171/GEN.

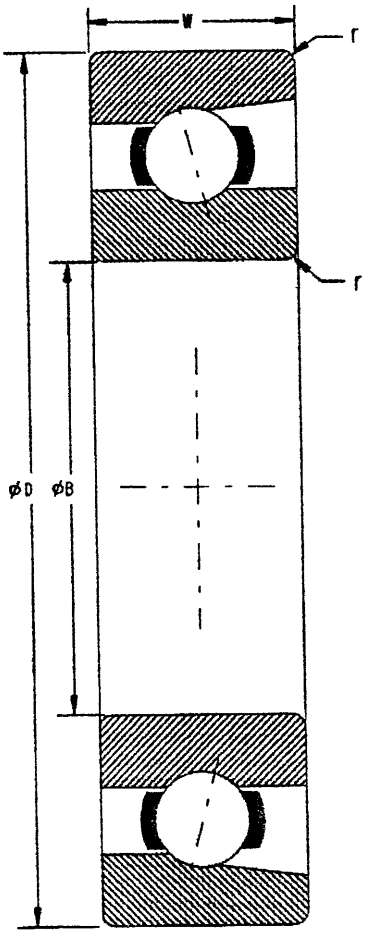


FIGURE 1.
OPEN (NO SHIELDS OR SEALS)
(CODE A)

BEARING CONFIGURATIONS

AMSC/NA

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REQUIREMENTS:

1. Boundary dimensions. The boundary dimensions shall be specified in the part number by the size code in table I.

2. Cages. Cages shall be manufactured from any material unless otherwise specified. Cages shall be specified in the part number by the codes in table II.

3. Shields and Seals. All angular contact bearings with counterbored outer rings shall be open (no shields or seals, see figure 1) and designated by the code "A" in the part number.

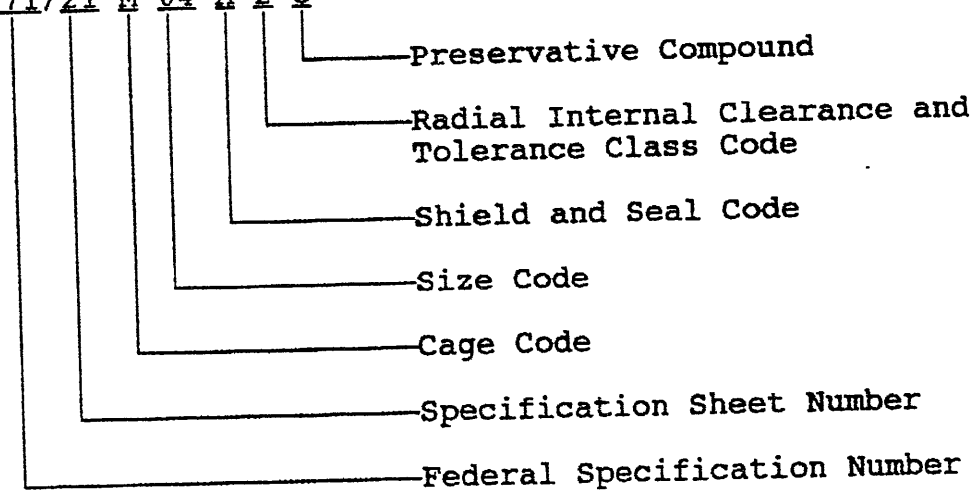
4. Radial Internal Clearance. The tolerance limits for radial internal clearance shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table III.

5. Tolerance Class. The tolerance limits for tolerance class ABEC-1 and ABEC-5 shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table III.

6. Preservative Compound. All bearings shall be supplied with preservative compound in accordance with MIL-P-197 and designated by the code "C" in the part number.

7. Part number. The part number consists of the letters FFB, the basic Federal specification number, the specification sheet number, and alpha or numeric codes for cage, size, shields and seals, radial internal clearance and tolerance class, and preservative compound.

EXAMPLE: FFB171/21 M 04 A 2 C



FFB171/21M04A2C indicates - Bearing, cage-machined-bronze or brass: bore 20mm; outside diameter 47mm: width 14mm: open (no shields or seals); radial internal clearance symbol 0 (normal): tolerance class ABEC-1; preservative compound in accordance with MIL-P-197.

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TABLE I. Boundary dimensions.

Size Code 1/	OB Bore	OD Outside Diameter	W Width	Chamfer $r_s$min 2/	Shaft Shoulder Diameter Min 3/	C Dynamic Load Rating (Pounds)
00	10	30	9	0.6	15	1160
01	12	32	10	0.6	17	1310
02	15	35	11	0.6	20	1555
03	17	40	12	0.6	22	1925
04	20	47	14	1.0	26	2200
05	25	52	15	1.0	31	2660
06	30	62	16	1.0	36	2925
07	35	72	17	1.1	42	4070
08	40	80	18	1.1	47	5390
09	45	85	19	1.1	52	5355
10	50	90	20	1.1	57	5805
11	55	100	21	1.5	63.5	8470
12	60	110	22	1.5	68.5	9185
13	65	120	23	1.5	73.5	11125
14	70	125	24	1.5	78.5	11970
15	75	130	25	1.5	83.5	13745
16	80	140	26	2.0	90	13095
17	85	150	28	2.0	95	15990
18	90	160	30	2.0	100	20450
19	95	170	32	2.1	107	21180
20	100	180	34	2.1	112	24015
21	105	190	36	2.1	117	25935
22	110	200	38	2.1	122	28260
24	120	215	40	2.1	132	30560
26	130	230	40	3.0	144	24260
28	140	250	42	3.0	154	36695
30	150	270	45	3.0	164	42030
32	160	290	48	3.0	174	47565
34	170	310	52	4.0	188	51175
36	180	320	52	4.0	198	53065
38	190	340	55	4.0	208	57980
40	200	360	58	4.0	218	60870

1/ Size codes in Table I are industry bore codes.

2/ The minimum bearing chamfer ($r_s$min) must clear the maximum shaft or housing fillet radius. This dimension does not control the shape of the bearing chamfer contour.

3/ For reference purposes only. Maximum shoulder heights shall be such as to permit bearing removal pressure to be applied to the bearing inner ring. In cases where this requirement cannot be met, provisions shall be made for the removal of the bearing from the shaft without damage to the bearing or other parts.

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TABLE II. Codes for Cages.

Code	Cage
J	Pressed steel or molded plastic
M	Machined bronze or brass
T	Machined non-metallic (phenolic)
Y	Pressed brass

TABLE III. Codes for Radial Internal Clearance and ABEC tolerance.

Code	Radial Internal Clearance	Tolerance Class
1	Symbol 2 (Less than normal)	ABEC-1
2	Symbol 0 (Normal)	
3	Symbol 3 (Greater than normal)	
4	Symbol 4 (Greater than symbol 3)	
5	Symbol 2 (Less than normal)	ABEC-5
6	Symbol 0 (Normal)	
7	Symbol 3 (Greater than normal)	
8	Symbol 4 (Greater than symbol 3)	

MILITARY INTERESTS:

Military Coordinating Activity:
DLA - IS

Custodians:
Army - AT
Air Force - 11

Review Activities:
Army - AR, EA, MI
Navy - OS

User Activity:
Army - GL, ME

CIVIL AGENCY COORDINATING ACTIVITIES:
GSA-FSS

PREPARING ACTIVITY:
DLA-IS

(Project 3110-0848-21)

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